The dimension of entirely ideal right-angled hyperbolic polytopes

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Abstract

We prove that an entirely ideal finite-volume right-angled hyperbolic polytope has dimension at most four. Fix a cusp and choose a facet whose outward unit normal has maximum cusp height. At the actual projection of the cusp to that facet, the neighboring walls determine a spherical 60-degree code. In dimension five, Musin's four-dimensional kissing-number theorem and Kolpakov's minimal-facet theorem force this facet to be a regular ideal 24-cell. The metric equality case makes every adjacent facet have the same maximum height. Connectivity then contradicts the zero-height facets at the original cusp. The regular ideal right-angled 24-cell attains the bound.

1 The two numerical inputs

A hyperbolic polytope is *entirely ideal* when all its vertices are ideal. Throughout, polytopes have finitely many facets and finite volume. Right-angled means that every actual dihedral angle is $\pi/2$. No arithmetic hypothesis is imposed.

Theorem 1. *An entirely ideal finite-volume right-angled hyperbolic polytope has dimension at most four. Equality is attained by the regular ideal right-angled* 24*-cell.*

We isolate the two substantive published theorems used in the proof. The first is a spherical-code cardinality bound; the second includes a metric equality statement.

Proposition 2 (Musin)**.** *If distinct vectors in a finite subset of $S^3 \subset \mathbb{R}^4$ have pairwise inner products at most* $1/2$*, the subset has at most* 24 *members.*

This is the Main theorem of [2, p. 4]. It is equivalent to the four-dimensional kissing number being 24. We use no uniqueness assertion for the kissing configuration.

Proposition 3 (Kolpakov)**.** *An entirely ideal right-angled hyperbolic four-polytope has at least* 24 *facets. If it has exactly* 24*, it is isometric to the regular ideal right-angled* 24*-cell.*

This is Theorem 4 of the published article [1, p. 1945]. Its equality conclusion will identify a hyperbolic polytope, rather than identify a spherical kissing arrangement. This distinction lets us treat equality with an elementary calculation in the standard 24-cell model.

2 Cusp height and an actual projection point

Use the Lorentz model of $\mathbb{H}^n$ in $\mathbb{R}^{n,1}$, with signature $(-,+,\dots,+)$ and future unit timelike points. For each actual facet choose its outward *unit* spacelike normal m. Thus

$$P = \{x \in \mathbb{H}^n : \langle x, m\rangle \le 0 \text{ for every actual facet normal } m\}.$$

Unit normalization will be retained for every comparison of heights.

We use the standard all-wall acuteness property

$$\langle m, m' \rangle \leq 0 \qquad (m \neq m'). \tag{1}$$

It applies also when the two facets do not share a ridge. As recalled in [3, p. 42, §2.1], the nonobtuse ridge-angle condition implies that nonadjacent supporting hyperplanes do not intersect, and the outward-normal Gram matrix has nonpositive off-diagonal entries. In particular, (1) is not an extra condition on unobserved intersections of supporting walls.

Fix an ideal vertex and a future null representative v. For every facet normal put

$$H(m) = -\langle v, m \rangle \geq 0. \tag{2}$$

The incident facets have height zero. Some facet has positive height, since a vertex of a full-dimensional finite-volume polytope is not incident to every facet. Let n have maximum height $H_0 > 0$.

Lemma 4 (Maximum-height facet)**.** *The point*

$$q = \frac{v}{H_0} + n \tag{3}$$

is an actual point of the facet $Q = P \cap n^{\perp}$. For every facet normal m_i adjacent to n across a ridge,

$$t_i = -\langle q, m_i \rangle = \frac{H(m_i)}{H_0} \in [0, 1]. \tag{4}$$

Proof. Since $\langle v, n \rangle = -H_0$ and $n^2 = 1$, direct products give

$$q^2 = -1, \qquad \langle q, n \rangle = 0, \qquad \langle v, q \rangle = -H_0 < 0.$$

Thus q is future unit timelike and belongs to $n^{\perp}$. For any other actual normal m, (1) gives

$$\langle q, m \rangle = -\frac{H(m)}{H_0} + \langle n, m \rangle \leq 0. \tag{5}$$

Hence it lies in all the halfspaces, so is an actual point of Q. Right angles give $\langle n, m_i \rangle = 0$ for a ridge neighbor. Formula (4) follows from maximality of H_0. □

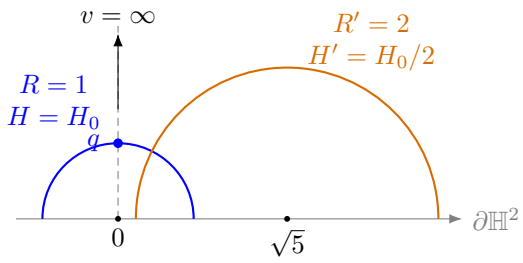
The point q may initially lie on the boundary of Q. For example, a neighboring wall through the chosen cusp has $t_i = 0$. The proof does not assume that q is interior; equality will force that later. Checking (5) against every wall is what makes this a projection into the actual facet, rather than merely a point on its supporting hyperplane.

In an upper half-space model with the chosen cusp at infinity, nonvertical facet walls are hemispheres. Their unit-normal heights are proportional to the inverse radii. Thus a maximum-height facet is a smallest-radius hemisphere, and q is its top point. This interpretation is illustrated in Figure 1.

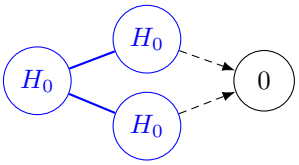
3 A spherical code in the selected facet

Every facet of Q comes from exactly one actual ridge neighbor m_i of n. Because the dihedral angle is right, $m_i \in n^{\perp}$ and remains a unit vector for the intrinsic Lorentz form. Use q as the time axis in this space. There are unit spatial vectors u_i such that

$$m_i = \big(t_i, \sqrt{1 + t_i^2}\, u_i\big), \qquad u_i \in S^{n-2}. \tag{6}$$



Orthogonal walls:
$(\sqrt{5})^2 = 1^2 + 2^2, \quad \sinh d(q, m') = 1/2.$



Equality propagates H_0 across every ridge. A connected path to a cusp facet is impossible.

Figure 1: The left panel is a genuine upper half-plane section of two orthogonal hemispherical walls, with the indicated radii. The right panel records propagation on the facet graph; it makes no metric assertion about a planar drawing of that graph.

Lemma 5 (The code bound)**.** *Distinct neighboring walls give distinct u_i, and*

$$u_i \cdot u_j \le \frac{t_i t_j}{\sqrt{(1+t_i^2)(1+t_j^2)}} \le \frac{1}{2}. \tag{7}$$

In dimension five, Q has at most 24 facets.

Proof. Apply (1) to the two actual normals m_i, m_j and substitute (6). Since $t_i, t_j \in [0,1]$, each factor $t_i/\sqrt{1+t_i^2}$ is at most $1/\sqrt{2}$, proving the second inequality. If $u_i = u_j$, then

$$\langle m_i, m_j \rangle = -t_i t_j + \sqrt{(1+t_i^2)(1+t_j^2)} > 0,$$

contrary to (1). Thus this is a code with as many members as facets of Q. For $n = 5$ it lies in $S^3 \subset \mathbb{R}^4$; Proposition 2 gives the asserted bound. □

Suppose now that P is entirely ideal and $n = 5$. The four-facet Q inherits entire ideality, finite volume and right angles. These inheritance statements are used for a face of dimension four, not for ideal polygon angles. Proposition 3 and Lemma 5 give

$$24 \le \#\{\text{facets of } Q\} \le 24. \tag{8}$$

Consequently Q is a regular ideal right-angled 24-cell.

4 The equality point and global propagation

We need a little more than the facet count. All neighbors must have the same height, which follows from the metric equality case of Proposition 3.

Lemma 6 (The center is forced)**.** *In a regular ideal right-angled 24-cell, a future unit timelike point whose products with all outward unit normals have negatives at most one is the center. All those negatives then equal one.*

Proof. In signature $(-,+,+,+,+)$ use the standard outward unit normals

$$(1, \epsilon e_i + \delta e_j), \qquad 1 \le i < j \le 4, \quad \epsilon, \delta \in \{1, -1\}. \tag{9}$$

The center is $(1,0)$, and each normal has an opposite spatial partner: $(1,r)$ and $(1,-r)$. Write the given point as (a,w). Then $a > 0$, $a^2 - \|w\|^2 = 1$, and the two negatives are $a - r \cdot w$ and $a + r \cdot w$. Their upper bounds by one imply $2a \le 2$. Future unit timelikeness implies $a \ge 1$. Thus $a = 1$ and $w = 0$, as required. □

Proof of Theorem 1. Assume first that P is an entirely ideal right-angled five-polytope. Fix the cusp representative v, height function and maximum H_0 as above. At any maximum-height facet the selected point q lies in the regular 24-cell Q, by (8). All neighboring heights satisfy $t_i \leq 1$. Lemma 6 forces q to be the center, so every $t_i = 1$. Formula (4) therefore gives

$$H(m_i) = H_0 \quad \text{for every facet adjacent to a maximum-height facet.}$$

The set of maximum-height facets is nonempty and closed under facet adjacency. The facet-adjacency graph is connected, so every facet would have height $H_0 > 0$. This contradicts the height-zero facets through the chosen cusp.

For completeness, connectivity here is the usual connectivity of the facet graph of the compact projective convex closure of P. This is a finite Euclidean polytope with the same face lattice. Its positive-dimensional ridges have hyperbolic relative interior, so the graph edges are precisely the actual ridge adjacencies used in the proof; cusp-only contacts do not serve as graph edges.

There is therefore no entirely ideal right-angled five-polytope. In every higher dimension a five-face would inherit finite volume, entire ideality and right angles, giving the same contradiction. The regular ideal right-angled 24-cell supplies the example in dimension four. □

The argument uses no field restriction, local search, numerical optimization or assumed classification of higher-dimensional polytopes. Its two numerical inputs are exactly Propositions 2 and 3. The all-wall check (5), the allowed zero heights, and the explicit opposite-normal calculation are included to keep projection and equality independent of any informal picture.

References

[1] A. Kolpakov, *On the optimality of the ideal right-angled* 24*-cell*, Algebraic & Geometric Topology **12** (2012), 1941–1960. doi:10.2140/agt.2012.12.1941.

[2] O. R. Musin, *The kissing number in four dimensions*, Annals of Mathematics (2) **168** (2008), 1–32. doi:10.4007/annals.2008.168.1.

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